SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI RAMAPURAM

Workload & Timetable Portal

Product Requirements Document (PRD) - Phase 1: Workload

Item	Detail
Document	PRD v0.2 (updated with your review answers)
Prepared	24 September 2026 (v0.1 issued earlier the same day; v0.2 adds your decisions)
Institution	SRM Institute of Science and Technology, Chennai Ramapuram - Faculty of Science and Humanities
Phase in scope	Phase 1 - Workload (preferences, allocation, workload details, class-wise matrix)
Phase deferred	Phase 2 - Timetable. Not built until you approve Phase 1. The architecture leaves room for it.
Sources analysed	Draft instructions (txt), Requirements v0.1 (docx), 5 curriculum PDFs, Data Science staff PDF, MCA staff xlsx, MCA workload xlsx, MCA class-wise matrix xlsx, B.Sc CS shift-wise staff list (docx), logo

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0. What changed in v0.2

Your answers to the review questions have been applied throughout this document. Nothing outside these changes was altered.

#	Decision	Effect on the design
1	Departments hold programmes. BCA Data Science is part of Computer Applications (changeable later).	Three departments: Computer Science (B.Sc CS, M.Sc CS), AIML (B.Sc AIML, M.Sc AIML), Computer Applications (BCA Data Science, BCA Gen AI, MCA, MCA Gen AI). Grouping is data, so it can be changed without code.
2	One grouped Programme dropdown replaces the separate UG/PG step.	Selector is now Department, then Programme (grouped under Undergraduate and Postgraduate), then Semester type. UG or PG is a property of the programme and stays visible in the banner.
3	Hour caps set as defaults, editable later: Assistant Professor 18, Associate Professor 16, Professor and Professor & HOD 12.	Stored as settings with a Settings screen. Exceeding a cap shows a warning, never blocks saving.
4	FTS (Full Time Scholar) data is not added for now.	FTS faculty are not seeded. Rows they taught appear as unallocated in the demo MCA cycle. Counting rules will be added when supplied.
5	Shift-2 details are left out for now.	Only Shift-1 staff are seeded. The data model keeps shift on the **section**, so per-shift repeats and per-shift splits can be added later as data.
6	Demo HOD login for every department.	Each department opens as a generic "Demo HOD". No real HOD names are used.
7	Workload PDF export is **mandatory**.	Promoted from Should to **Must**. The Export PDF button is on the finalized Workload Details page and must download a correct file. See W-09.
8	A sphere chart for **each semester** of each programme, not one overall.	Every semester card and every Workload Allocation page has its own sphere, calculated per programme and per semester.
9	The "three menus" are the options shown as soon as a semester is opened.	Renamed **Semester Workspace**. Three large option cards (Preference Management, Workload Allocation, Class-wise Matrix) are visible immediately and stay as a tab bar on every page inside the semester.

> **One thing to know about the faculty pool**
> A department can have several programmes and not every faculty member teaches every programme. To let the sphere reach 100%, the pool for each programme-semester is: faculty who submitted preferences there, plus anyone the HOD adds, with an "Add all department faculty" shortcut.

1. Purpose, scope and phasing

The Workload & Timetable Portal replaces manual Excel workload sheets and class-wise matrices with one live, data-driven web application. It works on desktops and mobile phones. This PRD covers **Phase 1 (Workload)** in full and only outlines Phase 2 (Timetable), as you instructed.

1.1 Goals

- Let each HOD run the whole workload cycle for their department from one workspace: review faculty preferences, allocate subjects, finalize, and view workload and the class-wise matrix.
- Let each faculty member submit and edit subject preferences per semester, then see the finalized "My Subjects".
- Derive the faculty workload and the class-wise matrix from **one source of truth** (the allocation table), so the two views can never disagree.
- Be configuration-driven: new departments, programmes, courses, faculty, sections and rules are added as data, not code.
- Never show a dead page: no 404, "request failed" or 500 in normal use. Every screen handles loading, empty and error states.

1.2 In scope (Phase 1)

- Demo login for HOD and Faculty (no credentials), then context selection (department, programme, semester type).
- HOD: Home, All Semesters, Workload Viewer, and a Semester Workspace with three always-visible options: Preference Management, Workload Allocation, Class-wise Matrix.
- Faculty: Home (semester cards, subject lists, preference submission and editing) and My Subjects.
- Progress sphere per semester, finalize gating, submitted/pending faculty views, editable allocations, audit trail, editable hour-cap settings, and **workload PDF export**.
- Seed data for the 7 departments/programmes you listed, with dummy data clearly flagged where real data is missing.
- Companion documents: this PRD and the How-To guide.

1.3 Out of scope (Phase 1)

- Timetable generation, rooms/labs, period timings, break rules, lab-allotment grids, elective clash handling (Phase 2).
- Real authentication, SSO, email invitations (demo login only for now).
- Shift-2 data and shift-specific screens; Full Time Scholar (FTS) data and counting rules.

> **Design decision**
> Timetable data (rooms, periods, timetable entries) is **not** built now, but tables and modules are named and reserved in section 14 so Phase 2 plugs in without touching the Phase 1 architecture.

2. What your files told us (data audit)

Every file in the reference folder was opened. The table shows what real data exists for each of your 7 departments and how the ERP will be seeded.

#	Department / programme	Type	Sem.	Curriculum	Faculty	Seed status
1	B.Sc Computer Science (Hons)	UG	8	PDF, Reg. 2024 (UCS codes). Two identical copies supplied.	DOCX: Shift-1 (15) seeded. Shift-2 and research scholars deferred	Real curriculum, real faculty
2	BCA Data Science	UG	8	PDF, Reg. 2024 (UDS codes)	PDF "Department of Data Science": 16 faculty (1 without ERP ID)	Real curriculum, real faculty
3	MCA	PG	4	PDF not supplied. Sem I and III subjects rebuilt from the workload sheet (PCA25 codes).	XLSX: 14 teaching staff, plus 3 B.Sc CS guests named in the workload sheet. FTS excluded	Partly real. Sem II and IV = dummy
3b	MCA GEN AI	PG	4	As above (PCA25 and PGI25 codes). Shares Sem I core with MCA.	Shared MCA pool	Partly real. Sem II and IV = dummy
4	BCA Gen AI	UG	8	PDF, Reg. 2024 (UGI and UDS codes)	None	Real curriculum, dummy faculty
5	M.Sc Computer Science	PG	4	PDF, Reg. 2025 (PCS25 codes). Two identical copies supplied.	None	Real curriculum, dummy faculty
6	M.Sc AIML	PG	4	None	None	All dummy
7	B.Sc AIML	UG	8	PDF, Reg. 2024 (UAI codes)	None	Real curriculum, dummy faculty

2.1 Findings that shape the design

- **Cross-department teaching is normal.** The MCA workload includes three B.Sc CS faculty (Dr R. Vanitha Mani, Mrs R. Sukanya, Ms P R Sukanya Sridevi) and one UG class ("II UG", Universal Human Values). Sukanya Sridevi also appears in the B.Sc CS Shift-2 list. So the allocation screen must let an HOD pick **any** faculty, including guests from other departments, and any class.
- **Faculty types beyond designations.** Professor and HOD, Professor, Assistant Professor, FTS (Full Time Scholar, excluded for now), and "AP/BSc(CS)" guests. The MCA staff sheet has no designation column; designations exist only in the workload sheet.
- **Workload hours are additive.** Each row in your workload sheet = theory hours + practical hours, and the faculty total is the sum (for example Dr K. Kalaiselvi: 5 + 3 + 2 = 10). Most Assistant Professors total exactly 18, which matches your default cap. Caps stay editable.
- **Labs have a Main and an IN-2.** Your class-wise sheet shows Java as 14 hours = 6 theory + 8 lab across two sections. That only balances if each lab session credits **both** the Main and the IN-2 with the lab hours (3 theory + 2 x 2 lab = 7 per section). Consistent with the workload sheet, where the same lab appears under two faculty. See rule BR-2 and question Q3.
- **Placeholders exist.** "NF1" and Data Science "Dr Jansi (New Faculty)" have no ERP ID (TINA and APPANNAN are FTS and are excluded for now). The system needs a "To Be Announced / New Faculty" placeholder that can later be replaced by a real record without breaking allocations.

- **Faculty with no workload row.** Dr J. Dhilipan and Dr R. Priya are on the MCA staff list but have no rows in the 2026-27 workload sheet. This is why the "No teaching this semester" option (P-10) matters for the Finalize gate.
- **Curriculum has slots, not always courses.** Semester tables contain "Minor Elective - I", "Multidisciplinary Course - II" and language options (Tamil/Hindi/French). The HOD must map each slot to an offered course for the semester before preferences open.
- **Sections and shifts.** UG uses sections A-D and PG uses A-B (your instruction). B.Sc CS also runs Shift-1 and Shift-2. Shift-2 is deferred, but shift is stored on the **section** so a later split or repeat per shift needs no redesign.
- **Academic year matters.** The B.Sc CS list is for 2025-26, the MCA workload for 2026-27. The portal will default to **2026-27** and store it on every cycle.

2.2 Data-quality issues found (to be handled gracefully)

Issue	Example in your files	How the ERP handles it
Name spelling variants	Sakthi Asvini / Aswini; Kottaisamy / Kottisamy; Uma Rani / Umarani	Faculty are keyed by ERP ID. Names are display-only. Seed loader normalises variants to one record.
Faculty in matrix or workload but not in staff list	Tinu Kumar, Tina, Appannan (FTS), Balaji; three emails in Sheet1 with no row	FTS are excluded for now. Others become flagged placeholder faculty, editable later.
Student-strength mismatch	II MCA A shown as 65 (DCN) and 63 (Cloud)	Strength stored per section and optionally per course offering. Seeder logs the conflict and uses the per-course value.
Code mismatch	Workload sheet lists UHV as UCD24E01T; curriculum says UCD24V02T	Course master is the curriculum. Unknown codes go to a "needs review" list instead of failing.
Unexplained totals	Matrix footer shows 72 / 40 / 32 and then 49 / 20 / 16	72 / 40 / 32 is reproduced by calculation. The second row is not derivable - see Q7.
Duplicate files	Two identical M.Sc CS PDFs; B.Sc CS PDF supplied twice	Treated as one source each.
Institution name differs	"Faculty of Science and Humanities" vs "Faculty of Liberal Arts and Business Studies"	Shown as a configurable header text.

3. Users, roles and demo login

Role	Responsibility in Phase 1	Status
HOD	Owns one department workspace: review preferences, allocate workload, finalize, view workload and matrices.	Core
Faculty	Submit and edit preferences per semester; view "My Subjects" after finalization.	Core
Admin / Dean / Deputy Dean	Institution-wide oversight, approvals (your workload sheet has Deputy Dean and Dean signature lines).	Reserved. Role model supports them; screens deferred.

3.1 Demo login (no credentials)

- Landing page shows two large buttons: **Continue as HOD** and **Continue as Faculty**.
- **HOD:** goes straight to context selection. The identity is a generic "Demo HOD" for the chosen department. Real HOD accounts are decided later.
- **Faculty:** because "My Subjects" and preferences belong to a person, the demo asks one extra question: a **"Who are you?" dropdown** listing seeded faculty (searchable, grouped by department). No password.
- A "Switch role" link is always in the header so you can move between HOD and Faculty during review.
- The login is isolated behind an auth service. Replacing it with ERP ID + institutional email (Requirements v0.1, section 4) later does not change any other screen.

> **Security note**
> Demo login gives full access to any role. It is for review only and must be replaced before real use. The API already checks a role and user id on every call, so real authentication is a swap, not a rewrite.

4. Context selection and navigation

4.1 Selection flow (both roles)

A department owns its programmes. UG or PG is a property of the programme, so there is no separate UG/PG step. The Programme dropdown is grouped by level.

Step	Dropdown	Example values	Notes
1	Department	Computer Science, AIML, Computer Applications	From the departments table.
2	Programme	Undergraduate: BCA Data Science, BCA Gen AI. Postgraduate: MCA, MCA Gen AI	Grouped by level. Auto-selected if the department has only one.
3	Semester type	Odd (1, 3, 5, 7) / Even (2, 4, 6, 8)	PG programmes show only their semesters, for example 1 and 3.
-	Academic year	2026-27 (default)	Small editable chip.

After step 3, the user enters the workspace. A permanent **context banner** shows department, programme (with UG or PG), semester type and academic year in bold on every page, with a "Change context" button. Changing context never loses saved data.

4.2 HOD sidebar

Menu	Content
Home	Welcome banner with context. One card per semester of the chosen type (for example Semesters 1, 3, 5, 7), each showing allocation progress, faculty submitted/pending and status.
All Semesters	The same semester cards for all semesters of the type, laid out as a grid with more detail (subjects count, sections, progress, finalization state).
Workload Viewer	Quick read-only view of the generated workload for each semester of the type. Semesters without a finalized workload show "No workload generated yet".

Clicking a semester card opens the **Semester Workspace**. Three large option cards are visible immediately: **Preference Management**, **Workload Allocation** and **Class-wise Matrix**. Each opens its page, and the same three options stay in a tab bar on every page inside the semester so the HOD never has to go back to reach another one. Every page has a Back button (returns to the previous page) and the sidebar stays available. On mobile the sidebar becomes a slide-out drawer with a bottom quick bar.

4.3 Faculty sidebar

Menu	Content
Home	Welcome banner with context. One card per semester of the chosen type with the published subject list (name, code, category, L-T-P-C). Faculty select subjects, submit, edit and resubmit here.
My Subjects	Finalized allocation per semester: subject, code, category, class and section, role (Main / IN-2), theory and practical hours, total workload.

5. Functional requirements - HOD

Priority: **M** = Must, **S** = Should, **C** = Could. Each requirement is testable.

5.1 Home, All Semesters, Workload Viewer

ID	Requirement	Pri.
H-01	Home shows a welcome message and the bold context banner (department, programme with UG/PG, semester type, academic year).	M
H-02	Home lists **all** semesters of the selected type as cards, each with: semester number, subject count, faculty submitted / pending, a small **sphere with allocation progress %** for that semester, state (Not started, In progress, Finalized).	M
H-03	All Semesters shows the same data for every semester of the type in a denser grid. Semester cards open the Semester Workspace.	M
H-04	Workload Viewer shows one panel per semester. Finalized semesters show the workload matrix summary; others show "No workload generated yet".	M
H-05	Every page has Back, and a link to Change context.	M

5.2 Preference Management (tab 1)

ID	Requirement	Pri.
P-01	Lists every faculty in the department pool with name, designation, ERP ID, and a status chip: Not submitted, Submitted, Allocated.	M
P-02	Header tiles: **Submitted (n)** and **Pending (n)**. Clicking a tile filters the list and shows those faculty with details (name, ERP ID, designation, email, phone).	M
P-03	Clicking a faculty opens their detail panel with all preferred subjects: name, code, category (Core / Discipline Elective / Minor / Multidisciplinary / Skill / Ability / Value Added / Project), L-T-P-C.	M
P-04	For each preferred subject the HOD sets a review status: **Approve, Keep Pending, Deny** (Requirements v0.1, section 5.1), or removes it.	M
P-05	The HOD can **add a subject the faculty did not prefer** (search dropdown of the semester curriculum). It is tagged "HOD-offered".	M
P-06	For each subject to be allocated the HOD enters: component (Theory / Lab / Theory + Lab), theory hours, practical hours, **class**, **section**, and **role** (Main In-charge / Assistant In-charge IN-2). Hours default from the course L-T-P and remain editable.	M
P-07	One subject may be given several rows (for example one faculty as Main for Section A and IN-2 for Section B).	M
P-08	Live validation on the panel: duplicate Main, missing class/section, hours outside curriculum range (warning), cap exceeded (warning, once caps are configured). Errors block submission, warnings do not.	M
P-09	**Submit Allocation** saves that faculty's allocation, marks them Allocated, and updates the sphere, Workload Allocation tab and Class-wise Matrix immediately.	M
P-10	A faculty can also be marked **No teaching this semester** (with reason) so Finalize is not blocked by leave or admin roles. (Q5)	S
P-11	Reviewing pending faculty is possible: HOD can allocate directly without a submitted preference.	M

5.3 Workload Allocation (tab 2) and finalization

ID	Requirement	Pri.
W-01	Shows one row per faculty with all allocated subjects and computed totals (theory, practical, total).	M

ID	Requirement	Pri.
W-02	A **spherical progress chart for this programme and semester** fills as faculty are allocated: allocated faculty / faculty in the pool. 100% shows a full sphere. Each semester has its own sphere; nothing is aggregated across semesters or programmes.	M
W-03	Every row is editable (Edit, Save, Cancel). Edits recompute totals, the sphere, and the matrix at once.	M
W-04	**Finalize Workload** at the bottom stays disabled until every faculty in the pool is Allocated (or marked No teaching). A tooltip and inline text state what is missing.	M
W-05	Coverage warning before finalizing: any subject-class-section with no Main in-charge is listed (soft check, HOD may still finalize).	S
W-06	After finalizing, a **Continue to Workload Details** button appears.	M
W-07	**Workload Details page:** top tiles Faculty Submitted / Faculty Pending (clickable, showing the faculty and their details); below, a clean matrix of all faculty with their allocations and totals.	M
W-08	Finalized workload can be reopened for editing. Reopen creates a version snapshot and audit entry; nothing is silently overwritten.	M
W-09	**Export PDF (mandatory).** An Export PDF button on the finalized Workload Details page downloads a PDF that must open correctly on desktop and phone. Contents follow your Excel sheet: institution and faculty header, department, programme, semester and academic year, the faculty workload table (S.No, Staff name and designation, Subject code, Course name, Class and section, No. of students, Theory, Practical, Hours) with totals, and signature lines Faculty In-charge / HOD / Deputy Dean Academic / Dean. The PDF is generated on the server from the same allocation data and is never a screenshot. It is disabled with an explanation until the semester is finalized. Long tables paginate with repeated headers.	M
W-10	**Hour caps.** A Settings screen edits caps per designation (defaults: Assistant Professor 18, Associate Professor 16, Professor and Professor & HOD 12). Totals show a warning when a cap is exceeded, counted across all programmes of the same semester type. Never blocks saving.	M

5.4 Class-wise Matrix (tab 3)

ID	Requirement	Pri.
M-01	Built automatically from allocations, following your MCA class-wise sheet (Appendix B): one block per class, columns per section with Main and Assistant, then total, theory and lab hours.	M
M-02	Updates the moment any faculty allocation is submitted or edited. No manual entry, no separate copy.	M
M-03	Unallocated subject-section cells are highlighted so gaps are obvious.	M
M-04	Class advisor per section is shown in the block header (for example "A SEC / Mrs M. Divya") and is editable.	S
M-05	Horizontal scroll with sticky subject column on mobile.	M

5.5 Cross-cutting HOD features

- **Dropdowns everywhere a choice is needed:** faculty, subject, class, section, role, component, status, semester.
- **Edit and save** on every action. Unsaved changes prompt before leaving.
- **Elective setup:** the HOD maps curriculum placeholders (for example Minor Elective - I) to actual offered courses per semester, and sets class-wide or combined-section delivery.
- **Faculty management (light):** add or edit faculty and guest faculty, including placeholders with no ERP ID.
- **Audit trail:** who changed what and when, visible per allocation.

6. Functional requirements - Faculty

ID	Requirement	Pri.
F-01	Same context selection as the HOD. Faculty default to their own department but may change it, since your data shows cross-department teaching.	M
F-02	Home welcomes the faculty by name and shows the bold context banner.	M
F-03	Home shows **all** semesters of the chosen type. Each card lists the published subjects: name, code, category, L-T-P-C.	M
F-04	Faculty tick the subjects they prefer per semester using checkboxes or a dropdown, then **Submit Preferences** to the HOD of that department.	M
F-05	Faculty can edit, modify and resubmit preferences at any time until the HOD finalizes that semester. After finalization the semester is read-only with a clear notice.	M
F-06	Preferences show status per subject as set by the HOD (Submitted, Approved, Pending, Denied) so faculty know where they stand.	S
F-07	**My Subjects** shows the finalized allocation per semester: subject, code, category, class, section, role, theory and practical hours, semester total.	M
F-08	Every submission appears immediately in the HOD login for that department and semester.	M
F-09	Empty states are friendly: "Nothing allocated yet - your HOD is still working on this semester".	M

7. Business rules and calculations

ID	Rule	Evidence / note
BR-1	**Workload hours** for a faculty = sum of (theory hours + practical hours) over all their allocations in the cycle. Always computed, never typed.	Kalaiselvi: DCN 3+2, AWT 3, DDPS 2 = 10 (matches sheet).
BR-2	**Lab double credit.** A lab has a Main and an IN-2; each is credited the lab hours. Class-matrix "Lab" = sum of lab hours across Main and IN-2; "Theory" = sum of theory hours of the Main.	Java 3 theory + 2 lab x 2 faculty = 7 per section, 14 for two sections (6 + 8). Matches sheet. Confirm - Q3.
BR-3	Roles: **Main In-charge** and **Assistant In-charge (IN-2)**. Theory Main and Lab Main may be different people.	DSA is theory-only. Java lab has a separate IN-2.
BR-4	Default hours come from the course L-T-P. Editable with an optional reason. Outside-range values raise a warning.	PG Java L3 T0 P2 = 3 theory + 2 practical, same as sheet.
BR-5	Only one **Main** per subject + class + section + component.	Validation error.
BR-6	Sections: UG A, B, C, D. PG A, B. Configurable per class. Each section may carry a shift, so shifts can repeat or split per department later. Shift-2 is deferred.	Your instruction and shift list.
BR-7	Odd = semesters 1, 3, 5, 7. Even = 2, 4, 6, 8. PG has 4 semesters: odd = 1, 3, even = 2, 4.	Your draft.
BR-8	Faculty pool for a programme-semester = faculty who submitted preferences there + faculty the HOD adds (own department or guests), with an "Add all department faculty" shortcut.	Lets the sphere reach 100% when a department runs several programmes.
BR-9	Progress % = faculty Allocated (or No teaching) / faculty in pool x 100, calculated **separately for each programme and semester**.	One sphere per semester.
BR-10	Finalize enabled only at 100%. Reopen allowed; it creates a version and returns the state to In progress.	Your draft and Requirements section 8.
BR-11	Elective and multidisciplinary placeholders must be mapped to real courses before appearing to faculty.	Placeholder rows in every PDF.
BR-12	Subject categories: Core (C), Discipline Elective (D), Minor Elective (E), Multidisciplinary (M), Ability Enhancement (A), Skill Enhancement (S), Value Added (V), Project/Internship (P), Mandatory (Y). PG adds Professional Core, Generic Elective (G), Ability (AE).	From the curriculum PDFs.
BR-13	Caps per designation are **editable settings**. Defaults: Assistant Professor 18, Associate Professor 16, Professor and Professor & HOD 12. A cap is checked on the faculty's total across all programmes in the semester type. Warning only.	Your answer; "just for now".
BR-14	Placeholder faculty (no ERP ID) can be allocated and later merged into a real faculty record. FTS faculty are excluded until counting rules are supplied.	NF1, New Faculty.
BR-15	Every create, edit, finalize and reopen writes an audit row (user, time, before, after).	Requirements section 8.

8. Data model

PostgreSQL, normalised. **allocation** is the single source of truth. Workload totals, progress and the class-wise matrix are computed from it (database views or service queries), never stored as duplicates.

Table	Key columns	Purpose
department	id, name, code	Computer Science, AIML, Computer Applications.
programme	id, department_id, name, level (UG/PG), total_semesters, regulation	B.Sc CS, BCA DS, MCA, MCA Gen AI ... Level drives the grouped dropdown.
academic_year	id, label, is_current	2026-27.
course	id, code, title, category, L, T, P, credits, offering_department_id	Course master. Shared courses (for example minors) are stored once.
curriculum_slot	id, programme_id, semester_no, course_id (nullable), slot_label, category	Semester structure, including placeholder slots.
elective_offering	id, slot_id, course_id, academic_year_id, delivery (class-wide / combined)	Maps a placeholder to a real course.
class_group	id, programme_id, year_of_study, semester_no, academic_year_id	For example "II MCA GEN AI".
section	id, class_group_id, name, shift (nullable), strength, class_advisor_id	A-D UG, A-B PG. Shift lives here.
faculty	id, erp_id (nullable), name, designation, email, phone, home_department_id, faculty_type, is_placeholder, is_dummy	One person, one row. Guests share the table.
workload_cycle	id, department_id, programme_id, academic_year_id, semester_type, semester_no, status (not_started / in_progress / finalized), version	One cycle per semester context.
preference_submission	id, cycle_id, faculty_id, status, submitted_at, updated_at	A faculty's submission for a semester.
preference_item	id, submission_id, course_id, review_status, remark	Each preferred subject and its review status.
allocation	id, cycle_id, faculty_id, course_id, section_id (or class-wide), role, component, theory_hours, practical_hours, source (preferred / hod_offered), created_by	**Source of truth.**
faculty_cycle_state	cycle_id, faculty_id, state (pending / allocated / no_teaching), reason	Drives the sphere and Finalize gating.
workload_version	id, cycle_id, version_no, snapshot_json, created_at, reason	Snapshots on finalize and reopen.
audit_log	id, user_ref, entity, entity_id, action, before_json, after_json, at	Traceability.
policy_config	id, key, scope, value	Hour caps per designation (18 / 16 / 12 defaults), default sections, labels. Edited on the Settings screen.
export_log	id, cycle_id, format, generated_at, generated_by, version_no	Records each PDF export.

Derived views: **v_faculty_workload** (totals per faculty per cycle), **v_class_matrix** (per class, section, course: Main, Assistant, theory, lab, total) and **v_cycle_progress**.

9. Technical architecture and coding standards

Built to your university standard.

Layer	Choice	Notes
Backend	Python 3.14, FastAPI	Pydantic v2 request and response models. Async endpoints.
Database	PostgreSQL	SQLAlchemy 2.x ORM and Alembic migrations (proposed). Seed scripts idempotent.
PDF export	Server-side PDF generation (Python, for example ReportLab)	Library to be confirmed on Python 3.14 in M1. Generated from allocation data.
Frontend	React with TypeScript	Vite build. Responsive for desktop and mobile.
HTTP client	axios only	No TanStack. State via React hooks and context.
Typing rule	No `any`	Strict tsconfig, ESLint rule `no-explicit-any` set to error. `interface` for every request and response.

Risk to verify on day one

Python 3.14 is recent. Before coding I will pin and test FastAPI, Pydantic, SQLAlchemy and the PostgreSQL driver on 3.14. If any package lacks support, I will tell you the exact package and propose the smallest fix rather than silently downgrading.

9.1 Backend folder structure

Folder	Responsibility
app/controllers/	REST endpoints only (FastAPI routers). Validate input, call a service, return a response model. No SQL, no business logic.
app/services/	Business logic: workload calculation, validation rules, finalize and reopen, matrix building. Gets data through repositories and modifies it.
app/repository/	All database operations (queries, inserts, updates). The only layer that touches the session.
app/models/	SQLAlchemy entities.
app/schemas/	Pydantic request and response models.
app/core/	Config, database session, error handling, logging.
app/seed/	Seed loaders and the JSON/CSV source data, with dummy records flagged.
alembic/ and tests/	Migrations. Unit tests for services, API tests for controllers.

9.2 Frontend folder structure

Folder	Responsibility
src/api/	axios instance and one module per resource (context, faculty, preferences, allocations, workload, matrix). Only place that calls the backend.
src/components/	Feature UI components, each in its own file: layout shell, sidebar, context banner, semester card, preference panel, allocation form, matrix table, faculty views.
src/common/	Reusable building blocks: Button, Select, Modal, DataTable, Toast, EmptyState, Skeleton, ProgressSphere.
src/utils/	Utilities: hour calculation, formatters, semester-parity helpers, validators.
src/interfaces/	TypeScript interfaces for all requests, responses and props. **Added** so no inline or `any` types are needed.

9.3 Reliability by design

- Every route the UI links to exists. A catch-all route shows a friendly "Take me home" page instead of a 404.
- The API returns one consistent error shape (code, message, details). The UI maps it to a readable message, never a raw stack trace.
- Every screen has loading, empty, error and retry states.
- Database constraints back the validation rules (unique Main per subject-section-component, foreign keys).
- Automated tests cover the calculation rules in section 7 using your MCA sheet as a fixture.

10. REST API outline

Base path `/api/v1`. Names are indicative and will be finalized in the API contract with the code.

Area	Endpoints	Used by
Auth (demo)	POST /auth/demo (role, faculty_id optional); GET /auth/me	Landing
Context	GET /departments; GET /departments/{id}/programmes?level=; GET /academic-years; GET /programmes/{id}/semesters?type=	Selector
Curriculum	GET /programmes/{id}/semesters/{n}/courses; GET/PUT /elective-offerings	Faculty home, HOD
Faculty	GET /faculty?department_id=; POST /faculty (incl. guest and placeholder); PUT /faculty/{id}	HOD
Cycles	GET /cycles?context; GET /cycles/{id}/progress; POST /cycles/{id}/finalize; POST /cycles/{id}/reopen	HOD home, allocation
Preferences	PUT /cycles/{id}/preferences/{faculty_id}; GET /cycles/{id}/preferences (submitted / pending); PATCH /preference-items/{id}/status	Faculty, HOD
Allocations	GET /cycles/{id}/faculty/{fid}/allocations; PUT (replace set) and POST /submit; DELETE /allocations/{id}	HOD
Workload	GET /cycles/{id}/workload; GET /workload/summary?context (Workload Viewer); GET /faculty/{fid}/my-subjects	HOD, Faculty
Matrix	GET /cycles/{id}/class-matrix	HOD
Export and settings	GET /cycles/{id}/workload/export.pdf; GET/PUT /settings/caps	HOD
Audit	GET /audit?entity=&entity_id=	HOD

11. UI/UX and responsive design

- **Branding:** your SRM Chennai Ramapuram logo sits top-left in the header on every page. Palette from the logo: SRM blue and gold on a white surface.
- **Typography:** large, high-contrast sans-serif (minimum 16 px body on mobile), clear hierarchy, bold context banner.
- **Welcoming and clear:** friendly greeting on each Home, a "You are here" banner, breadcrumbs plus Back.
- **Progress sphere:** animated liquid-fill sphere (SVG), percentage in the centre, colour shifts as it approaches 100%.
- **Mobile:** sidebar becomes a drawer; faculty lists become cards; matrices scroll horizontally with a sticky first column; touch targets at least 44 px; forms use full-width dropdowns.
- **Desktop:** persistent sidebar, split-panel preference review (faculty list left, details right).
- **Feedback:** toasts on save, inline validation, confirm dialogs for finalize and reopen, skeleton loaders.
- **Accessibility:** keyboard navigation, labelled controls, colour is never the only signal.
- **Flag for dummy data:** a small "Sample data" badge on any dummy faculty, course or student count.

12. Seed data plan

Data	Source	Rule
Courses and semester structure	The five curriculum PDFs	Loaded as-is, including L-T-P-C and category. Placeholders kept as slots.
MCA and MCA Gen AI Sem I and III	Workload sheet (ODD 2026-27)	Courses, classes, sections, strengths, and the full allocation rebuilt as a demo cycle for validation.
MCA Sem II and IV	Not supplied	Dummy courses, flagged "Sample". Sem IV = Project Work.
Faculty	MCA xlsx, Data Science PDF, B.Sc CS DOCX (Shift-1), workload sheet	Real names, ERP IDs, emails, phones. Variants normalised. Missing ERP IDs = placeholders. FTS and Shift-2 not seeded.
Faculty for Depts 4, 5, 6, 7	None supplied	8 to 12 dummy faculty each, mixed designations, ERP IDs like DUMMY-001, flagged.
Sections	Your instruction	UG A, B, C, D. PG A, B. Strength: real for MCA, dummy 60 elsewhere.
Preferences	None supplied	A few dummy submissions per department so HOD screens are populated on first open. Clearly flagged.
HODs	Your answer	Generic "Demo HOD" for every department. Real HODs decided later.

Replacing dummy data later is done by re-running a seed import or editing in the UI. No code change.

13. Non-functional requirements

Area	Requirement
Devices	Latest Chrome, Edge, Safari, Firefox; phones from 360 px width; tablets; desktops.
Performance	Screens load in under 2 seconds on a normal connection with seed-sized data (hundreds of faculty, tens of classes). Matrix recomputes in under 1 second after a save.
Reliability	No 404, request-failed or 500 on any linked route in normal use. Graceful messages otherwise.
Data integrity	Foreign keys, unique constraints, transactions on Submit Allocation, Finalize and Reopen.
Security	Demo login for now. API structured for role checks. No secrets in the frontend.
Maintainability	Layered backend, typed frontend, no `any`, linted and formatted, README with setup steps.
Traceability	Audit log and versions retained. No silent overwrites.

14. Extensibility and Phase 2 readiness

- New department or programme = new rows plus curriculum import. No code change.
- New rule (caps, validations) = entries in policy_config and a small rule class in the services layer.
- New role (Admin, Dean) = role value plus new screens; existing endpoints already accept a role.
- **Reserved for Phase 2 (not built now):** room, period_config, break_config, faculty_availability, timetable_entry, synchronized_group (for electives and language subjects). They will reference the same cycle, class_group, section, course and allocation tables, so timetables are generated from finalized allocations exactly as your Requirements v0.1 describes.
- Timetable rules from Requirements v0.1 (Mon-Fri, break protection, lab continuity, max two lab hours per subject per day, and so on) are recorded and will be classified as hard or soft with you before Phase 2 starts.

15. Delivery plan and acceptance criteria

15.1 Milestones (Phase 1)

#	Milestone	Outcome
M0	PRD and How-To review (v0.2)	Decisions applied. Remaining questions in section 16 have defaults.
M1	Foundation	Repos, Python 3.14 compatibility check, PostgreSQL schema, migrations, seed data for 7 departments.
M2	Context and shells	Demo login, selector, HOD and Faculty layouts, sidebars, banner, responsive shell.
M3	Faculty preferences	Semester cards, subject lists, submit, edit, resubmit, status display.
M4	HOD allocation engine	Preference review, hours/class/section/role, validation, submit allocation, sphere, finalize gate.
M5	Workload details, matrix and PDF	Workload Details, Workload Viewer, Class-wise Matrix, My Subjects, **Export PDF**, caps Settings.
M6	Hardening	Tests, error and empty states, mobile QA, audit, performance pass.
M7	Your review	Sign-off. Only then does Phase 2 (Timetable) begin.

15.2 Acceptance checklist

1. Landing to workspace works for HOD and Faculty with no credentials.
2. All 7 departments can be selected and each opens with populated dropdowns.
3. Selecting Odd shows only odd semesters; Even shows only even. PG shows only 2 semesters per type.
4. Faculty submit, edit and resubmit preferences; the HOD sees them immediately.
5. HOD can review, approve, keep pending, deny, add a non-preferred subject, and set hours, class, section and role.
6. Submit Allocation updates the Workload Allocation tab, sphere and Class-wise Matrix at once.
7. Finalize is disabled until 100%, then enabled; Continue to Workload Details appears afterwards.
8. Submitted and Pending tiles open the correct faculty lists.
9. Rebuilt MCA demo cycle reproduces your sheet for non-FTS faculty (for example 10, 13, 16, 18). Rows taught by FTS show as unallocated, so matrix totals match once FTS are added.
10. Export PDF downloads a correct, readable PDF on desktop and phone, and its totals equal the on-screen totals.
11. Editing hour caps in Settings changes the warnings immediately.
12. Each semester card and Workload Allocation page shows its own sphere; sphere reaches 100% only for that semester.
13. Reopen after finalize creates a version and audit entry.
14. Every sidebar link, Back button and card opens a valid page. No 404, request failed or 500 found in a full click-through.
15. Layout usable on a 360 px phone and a desktop.
16. Code review: layered backend, `no-explicit-any` passes, only axios used.

16. Open questions and assumptions

Resolved items are marked. Anything open proceeds with the stated assumption.

#	Topic	Status
Q1	Department grouping	**Resolved.** Computer Science, AIML, Computer Applications (incl. BCA Data Science). Changeable later.
Q2	Academic year in context	Default 2026-27, editable.
Q3	Labs credit both Main and IN-2 with full lab hours (BR-2)	**Open.** Assumed yes from your sheet.
Q4	Hour caps	**Resolved for now.** 18 / 16 / 12, editable.
Q5	"No teaching this semester" option	Assumed yes.
Q6	FTS counting	**Deferred.** FTS not added for now.
Q7	Meaning of second matrix footer row (49 / 20 / 16)	**Open.** Not shown until clarified.
Q8	Electives and language subjects: class-wide or combined sections	**Open.** Assumed class-wide, one row per elective.
Q9	Shifts	**Deferred.** Shift-2 left out. Shift stored on section.
Q10	HOD identities	**Resolved.** Generic Demo HOD per department.
Q11	HOD allocating other-department faculty	Assumed yes. Guests' preferences optional.
Q12	Workload PDF export	**Resolved.** Mandatory (W-09).
Q13	Lab in-charge roles and lab-allotment grid	Assumed Phase 2.
Q14	Project, capstone, internship hours counted as workload	Assumed yes when entered as practical hours.
Q15	MCA / MCA Gen AI curriculum PDF and real data for other programmes	**Open.** Dummy data used, flagged.

16.1 Assumptions

- The "7 departments" are treated as 7 programmes (MCA and MCA Gen AI as two) under 3 departments.
- Tailwind CSS is used for styling and React Router for navigation (both allowed by your rules).
- Sample student strengths are dummy except for the MCA data.
- The Timetable phase will not be started until you say so.

Appendix A. Reference workload (MCA) rebuilt from your sheet

Totals below are recalculated from your "wl edit" workload sheet (ODD 2026-27) and will be used as a regression test: the ERP must reproduce them from its allocation data. The three FTS faculty (Tinu Kumar, Tina, Appannan; 9, 10 and 10 hours) are excluded for now.

Faculty	Designation	Rows	Total hours (sheet)
Dr K. Kalaiselvi	Professor and HOD	3	10
Dr R. Agusthiyar	Professor	4	13
Dr S. Uma Rani	Professor	3	16 (includes 6 h UG UHV)
Dr D. Kanchana	Assistant Professor	6	18
Dr M. Divya	Assistant Professor	6	18
Dr N. Krishnamoorthy	Assistant Professor	6	18
Dr S. Meenakshi	Assistant Professor	5	18
Dr M. Sowmya	Assistant Professor	5	18
Dr M. Sakthi Asvini	Assistant Professor	5	18
Dr K. Kottaisamy	Assistant Professor	5	18
Dr R. Sivasankari	Assistant Professor	6	18
Dr S. Chithra	Assistant Professor	6	18
Dr R. Vanitha Mani	AP, B.Sc CS (guest)	1	4
Mrs R. Sukanya	AP, B.Sc CS (guest)	1	4
Ms P R Sukanya Sridevi	AP, B.Sc CS (guest)	2	10

Worked example (Dr K. Kalaiselvi): Data Communication Networks II MCA A (Theory 3 + Practical 2 = 5) + Advanced Web Technology I MCA B (Theory 3) + Distributed Data Processing Systems II MCA GEN AI (Theory 2) = **10**.

Appendix B. Class-wise matrix layout

Derived from your "CW" sheet. The ERP renders one block per class (I MCA, I MCA GEN AI, II MCA, II MCA GEN AI).

Column	Content	Computed from
Class header	For example "I MCA GEN AI" with Section A / class advisor and Section B / class advisor	class_group, section.class_advisor
Subject	Short name (JAVA, DSA, DBT, ILP ...) and full name and code on hover	course
Section A: Main, Asst	Faculty names	allocation.role for section A
Section B: Main, Asst	Faculty names	allocation.role for section B
Total hours	Theory + lab for the row	BR-1, BR-2
Theory hours	Sum of theory hours	allocation.theory_hours
Lab hours	Sum of lab hours across Main and IN-2	allocation.practical_hours
Footer totals	Column totals per class (72 / 40 / 32 in your sample)	Aggregation

Sample check from your sheet, I MCA GEN AI: JAVA 14 (6 + 8), DSA 8 (8 + 0), DBT 14, ILP 14, DNN 8, GEN AI 14 = **72 total, 40 theory, 32 lab**. The ERP must reproduce these numbers once FTS data is added. Until then, DSA and other FTS-taught rows show as unallocated.

The lab-allotment grid at the bottom of your sheet (Lab 1, days x periods) belongs to the Timetable phase and is not built in Phase 1.